

where $\beta_{i,m} = \text{cov}(r_i, m)/\text{var}(m)$ is the beta of the asset with respect to the SDF. Equation (6.8) captures the “bad times” intuition that we had earlier from the CAPM. Remember that m is an index of bad times. The *higher* the payoff of the asset is in bad times (so the higher $\text{cov}(r_i, m)$ and the higher $\beta_{i,m}$), the *lower* the expected return of that asset. The higher beta in equation (6.8) is multiplied by the price of “bad times” risk, $\lambda_m = -\text{var}(m)/E(m)$, which is the inverse of factor risk, which is why there is a negative sign. Equation (6.8) states directly the intuition of Lesson 6 from the CAPM: higher covariances with bad times lead to *lower* risk premiums. Assets that pay off in bad times are valuable to hold, so prices for these assets are high and expected returns are low.

Just as the CAPM gives rise to assets having betas with respect to the market, multiple factors in the SDF in equation (6.4) gives rise to a multi-beta relation for an asset’s risk premium:

$$E(r_i) = r_f + \beta_{i,1}E(f_1) + \beta_{i,2}E(f_2) + \dots + \beta_{i,K}E(f_K), \quad (6.9)$$

where $\beta_{i,k}$ is the beta of asset i with respect to factor k and $E(f_k)$ is the risk premium of factor k . For macro factors, f_1 could be inflation and f_2 could be economic growth, for example. Bad times are characterized by times of high inflation, low economic growth, or both. For an example for multiple investment factors, f_1 could be the market portfolio and f_2 could be an investing strategy based on going long value stocks and short growth stocks. Value stocks outperform growth stocks in the long run (see chapter 7). Bad times are characterized by low market returns, value stocks underperforming growth stocks, or both.

4.3. MULTIFACTOR MODEL LESSONS

The key lessons in the multifactor world are in fact the same from the CAPM:

	CAPM (Market Factor)	Multifactor Models
Lesson 1	Diversification works. The market diversifies away idiosyncratic risk.	Diversification works. The tradeable version of a factor diversifies away idiosyncratic risk.
Lesson 2	Each investor has her own optimal exposure of the market portfolio.	Each investor has her own optimal exposure of each factor risk.
Lesson 3	The average investor holds the market.	The average investor holds the market.

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